



## Practice assignment 10 - Not Graded



This assignment will not be graded and is only for practice.

1 point

Match the functions of two variables in Column A with their directions in which the directional derivatives at (0,0) is maximized given in Column B.

☐

(1)  $\rightarrow$  (v)(1)  $\rightarrow$  (v)

☐

(1)  $\rightarrow$  (iii)(1)  $\rightarrow$  (iii)

☐

(2)  $\rightarrow$  (iv)(2)  $\rightarrow$  (iv)

☐

(3)  $\rightarrow$  (iv)(3)  $\rightarrow$  (iv)

☐

(2)  $\rightarrow$  (vi)(2)  $\rightarrow$  (vi)

☐

(4)  $\rightarrow$  (v)(4)  $\rightarrow$  (v)

☐

(3)  $\rightarrow$  (vi)(3)  $\rightarrow$  (vi)

☐

(4)  $\rightarrow$  (i)(4)  $\rightarrow$  (i)

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(1)  $\rightarrow$  (iii)(1)  $\rightarrow$  (iii)

(2)  $\rightarrow$  (iv)(2)  $\rightarrow$  (iv)

(4)  $\rightarrow$  (v)(4)  $\rightarrow$  (v)

1 point

Which of the following statements is(are) correct?

☐

Maximum directional derivative at (0,0) for the function  $f(x, y) = |x| + |y|$  is 1.

☐

The graph of the linear approximation to a function  $f(x, y)$  at a given point  $P$  is the tangent plane to  $f(x, y)$  at the point  $P$ .

☐

A tangent plane to a function  $f(x, y)$  exists at every point.

☐

All directional derivatives of a function  $f(x, y)$  at a point  $P$  are 0 if the tangent plane is parallel to the  $XY$ -plane.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

The graph of the linear approximation to a function  $f(x, y)$  at a given point  $P$  is the tangent



plane to  $f(x, y)$  at the point  $P$ .

All directional derivatives of a function  $f(x, y)$  at a point  $P$  are 0 if the tangent plane is parallel to the  $XY$ -plane.

1 point

Which of the following is(are) critical points of the scalar valued multivariable function

$$f(x, y) = 6x^2 - 2x^3 + 3y^2 - 6xy \quad f(x, y) = 6x^2 - 2x^3 + 3y^2 - 6xy?$$

☐ (1,1)

☐ (0,0)

☐ (1,-1)

☐ No critical points.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(1,1)

(0,0)

Consider two functions  $f_1, f_2$  defined as

$$f_1(x, y, z) = \ln(x^2 + y^2 + 2z^2) \quad f_1(x, y, z) = \ln(x^2 + y^2 + 2z^2)$$

and

$$f_2(x, y, z) = \tan^{-1}(x^2 + y^2 + 2z^2) \quad f_2(x, y, z) = \tan^{-1}(x^2 + y^2 + 2z^2).$$

If  $L_{f_1}(x, y, z)$  and  $L_{f_2}(x, y, z)$  are the linear approximation of  $f_1(x, y, z)$  and  $f_2(x, y, z)$  at point  $(1, -1, 1)$ , then find the value of

$$85(\nabla L_{f_1}(x, y, z) \cdot \nabla L_{f_2}(x, y, z)) \quad 85(\nabla L_{f_1}(x, y, z) \cdot \nabla L_{f_2}(x, y, z)).$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 30.0

1 point

$z = f(x, y) = -\frac{7}{6}x^2 + \frac{1}{3}y^2 + \frac{11}{6}z = f(x, y) = 6 - 7x^2 + 31y^2 + 611$  represents a surface. Let  $(x_0, y_0, z_0)$  be the point where the tangent plane to the given surface is parallel to the plane  $7x - 8y + 3z + 2 = 0$ . Find the value of  $11x_0 + 5y_0 + 4z_0$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 55

1 point

Find the number of points where the tangent plane to the function

$f(x, y) = x^3 + y^3 + 3x^2 - 9x - 12y + 20$  at the points is parallel to the  $XY$ -plane.

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 4

1 point

- Suppose  $ax + by + cz = d_1$  is a tangent plane to the function  $f(x, y) = x^3 + 3y^3 + 9xy$  at  $(-1, 2)$  and  $ax + by + cz = d_2$  is a tangent plane to the same function  $f(x, y)$  at  $(\alpha, 1)$ .

Use the above information to answer questions 7-8:

1 point

Which of the following options is(are) true?

☐

$$\alpha = -2$$

☐

$$\alpha = 2$$

☐

There are only two possible candidates for  $\alpha$ .

☐

$$|d_2 - d_1| = 12$$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

$$\alpha = 2$$

$$|d_2 - d_1| = 12$$

Find the value of  $\frac{2a+b+c}{d_1+d_2}$

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1

1 point

- Let  $L(x, y)$  be the linear approximation to  $f(x, y) = xe^y + \ln(x + y)$  at point  $(1, 0)$ .

Use the above information to answer questions 9-10:

Suppose  $L(x, y) = ax + by + c$ , where  $a, b, c \in \mathbb{R}$ . Find the value of  $a + b - c$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 5

1 point

Find the approximate value of the function  $f(x, y)$  at  $(1.1, 0.1)$  using the linear approximation function  $L(x, y)$ .

**No, the answer is incorrect.**

**Score: 0**

**Accepted Answers:**

(Type: Numeric) 1.4

1 point

[Check Answers](#)

Your score is: 0/10